

EXAM CODE: 2024049

BE-II EXAMINATION DEC 2023

SUB CODE: 3VLRG1

SUB. NAME: STRUCTURAL MECHANICS

BRANCH: CIVIL

DURATION: 3 HOURS

MAX. MARKS-60

Note: Total five questions are to be solved. There is an internal choice in each question. Each question carries equal marks. Any missing data may be assumed suitably and mentioned clearly

Q.1. Draw SFD & BMD of Beam for the beam as shown in figure-1. Also locate points of contraflexure if any.

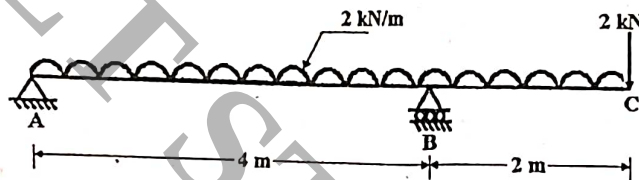


FIG - 1

OR

Draw SFD & BMD of Beam for the beam as shown in figure-2. Also locate points of contraflexure if any.

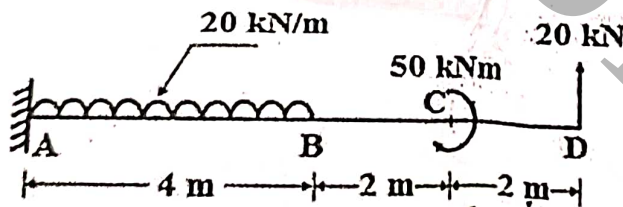
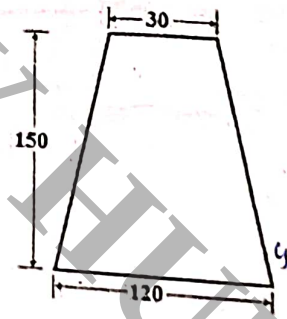


FIG - 2



All dimensions are in mm

FIG - 3

Q.2. A cast iron beam has a trapezium section as shown in figure-3. If tensile stress is not to exceed 100 N/mm^2 and compressive stress 80 N/mm^2 , what is maximum UDL the beam can carry safely over simply supported of span 4 m.

OR

Find the position and magnitude of maximum deflection for the beam as shown in figure-4. Take $E = 200 \text{ GPa}$ and $I = 4.3 \times 10^8 \text{ mm}^4$.

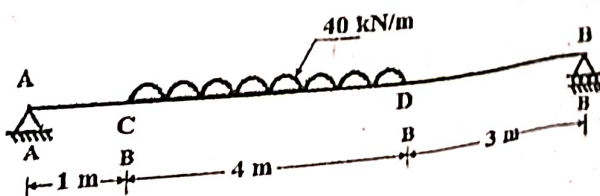


FIG - 4

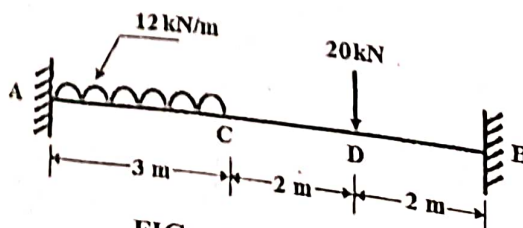


FIG - 5

POC of UDL
P. of zero SF = 2m
POC = 2m from A

$M_{max} = 2 \text{ kN-m}$
 $M_B = -8$

$y_t = 70, 80 = y_c$
 $I_{base} = 59.0625 \times 10^6$
 $I_{yx} = 3.9815 \times 10^6$
 $w_t = 2.5125$
 $w_c = 1.9683$

Q.3. Analyze the fixed beam as shown in figure-5 and draw its BMD.

OR

Free BMD: -22.5 , $40/48$
 $MA = -19.5361$, $-25.37352MB$

Analyse the following continuous beam as shown in figure-6 by "Three Moment Equation" and draw BMD.

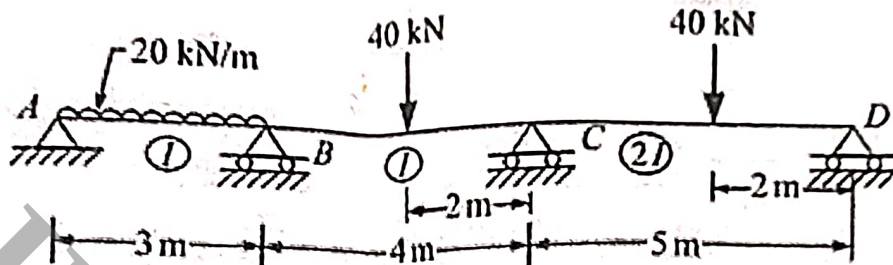


FIG - 6

Q.4. State the assumptions made in Euler's theory for column. Derive the expression for Euler's crippling load for a long column when both ends are fixed.

OR

$K=46.5$, $\lambda=129.03$ $P_c = 386.715 \text{ kN}$
 $P_c = 393.6771$
 $L = 1.71 \text{ m}$

Find the Euler's crushing load for a hollow cylindrical cast iron column, 15cm external diameter and 2cm thick, if it is 6m long and hinged at both ends. $E = 80 \text{ GPa}$. Compare this load with the crushing load as given by the Rankine's formula, using $\sigma_c = 550 \text{ MPa}$ and $a = 1/600$. For what length of the strut of this cross-section does the Euler's formula ceases to apply?

Q.5. A wooden beam of cross-section 100 mm x 150 mm as shown in figure-7 is used to carry an udl of 3 kN/m over an span of 4 m. Find the location of neutral axis. Also find the maximum tensile & compressive stresses developed in beam.

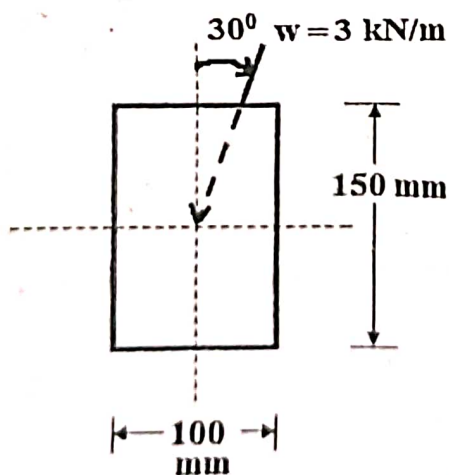


FIG - 7

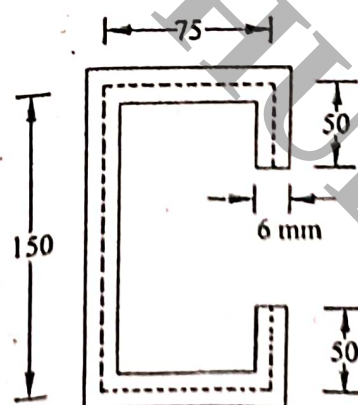


FIG - 8

OR

Locate the shear centre of the section as shown in figure-8. The thickness is 6 mm throughout.

$-3293 - \frac{2106}{83}$
 $\frac{166}{19.5361}$ 25.375